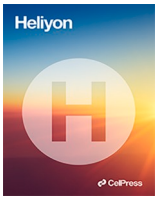
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Research article



Seroprevalence of transfusion transmissible infections (TTIS) among blood donors in SOS hospital Heliwa, Somalia

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ABSTRACT

Background: A transfusion-transmissible infection (TTI) refers to any infection that can be spread from one person to another through the injection of blood or blood products. The prevalence of these infections varies across countries, influenced by the disease burden within each population. To assess the severity of TTIs, the World Health Organization (WHO) has mandated pre-transfusion blood tests for Human Immunodeficiency Virus (HIV), Hepatitis B Virus (HBV), Hepatitis C Virus (HCV), and syphilis. This study aimed to determine the seroprevalence and trends of TTIs among blood donors at SOS Hospital in Mogadishu, Somalia, from 2016 to 2022.
Methods: A retrospective cross-sectional analysis was performed by examining SOS Hospital's blood bank records spanning from 2016 to 2022. The research included all blood donors screened for transfusion-transmissible infections (TTIs) during this period. Data obtained was input and analyzed utilizing Statistical Package for Social Science (SPSS) v.25.0 and Microsoft Excel 2010. Frequencies and percentages were calculated as part of the descriptive statistics. To analyze trends, Chi-square analysis was applied, and statistical significance between variables was determined using the p-value.
Results: There was a total of 36,296 people donated blood during study period. The majority of blood donors were males (99.8 %), primarily aged between 25 and 44 years (80.4 %) with family donors constituting 80.8 % of the participants. Among 36,296 donors, 1087 (2.99 %) tested positive for transfusion-transmissible infections (TTIs), declining from 4.27 % in 2016 to 1.98 % in 2022. The Chi-Square test confirmed a highly significant reduction in TTIs from 2016 to 2022 ($X2 = 57.625$, $p < 0.0001$), indicating a decreasing trend over the seven-year period.
Conclusion and recommendations: This study uncovers a moderate prevalence of transfusion-transmitted infections, indicating a notable decrease over time. Additionally, the findings underscore a gender disparity in blood donation, with replacement donors being predominant. It is imperative to conduct multi-center research endeavors to precisely identify the factors influencing transfusion-transmitted infections among blood donors.

1. Background

Blood transfusion is a crucial medical procedure that saves countless lives each year globally; it can be administered as whole blood to a single patient or processed into blood-derived products for use by multiple patients. However, it is known that blood transfusion

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can be associated with risks of transmitting certain infections [1,2] These infections are known as Transfusion Transmissible Infections (TTIs), which defined as any infection that is transmissible from person to-person through parenteral administration of blood or blood products. The magnitude of transfusion-transmitted infections (TTI) varies or differs from country to country depending on the burden of TTI's within that specific population. Measuring their severity, WHO (World Health Organization) has recommended pre-transfusion blood test for human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and syphilis as mandatory [3].

Transfusion transmissible infections are categorized into viral, bacterial, and parasitic types, with viral infections being the most common. Among these, the Human Immunodeficiency Virus (HIV) and Hepatitis B and C viruses are frequently responsible for diseases following transfusions. Prevalence of HBV infection varies significantly across different regions globally. The World Health Organization (WHO) has classified HBV prevalence into high endemicity (>8 %), intermediate (2–7 %) and low endemicity (<2 %) [4].

Blood safety continues to be a significant public health issue in sub-Saharan Africa due to deficiencies in national blood transfusion policies and services, lack of proper infrastructure, shortage of qualified personnel, and insufficient financial resources. Every country requires a safe, equitable, and sustainable blood supply to treat patients with severe bleeding from trauma or postpartum complications, children with severe anemia caused by malaria and malnutrition, individuals with inherited hemoglobin disorders, and those with bone marrow failure. However, blood for transfusion is also a potentially scarce resource with significant risks as a biological material, including transfusion-transmitted infections. Therefore, blood transfusion should be considered only as a last resort when no other options are available and when the benefits clearly outweigh the risks [5].

The discovery of transfusion-transmissible infections heralded a new era in blood transfusion practice worldwide with emphasis on two fundamental objectives, safety, and protection of human life. Blood safety remains an issue of major concern in transfusion medicine in East Africa e.g., Ethiopia [4]. Estimating the prevalence of TTIs, namely HBV, HCV, HIV and syphilis antibodies or antigen, among blood donors can uncover the presence of hidden infections in healthy-looking members of the general population and provide data that is important in formulating the strategies for improving the management of a safe blood supply. In addition, it can give us a guide to the magnitude of some sexually transmitted infections in the community in Heliwa district, Mogadishu-Somalia, the main source of blood donation is replacement donors and most of them are patient's relatives or friends. Proper screening of blood and selection of the donors is very important to insure a safe blood supply. It is mandatory to test each donor's blood for syphilis by a Venereal Disease Reference Laboratory (VDRL), and for HBsAg, anti-HCV, and anti-HIV.

On a global scale, an alarming pattern emerges wherein only 34 % of countries have implemented legislations pertaining to blood transfusion services, a trend conspicuously echoed in Africa. This deficiency is further exacerbated by inadequate financial backing, with a mere 68 % of nations allocating funds for blood banks. Particularly concerning is Africa's heavy reliance on direct budgetary allocations due to this financial shortfall. Furthermore, the absence of robust inspection and licensing systems poses a substantial threat to the safety and quality of transfusion services. Additionally, significant disparities persist in blood collection rates, demonstrating notable variations among diverse income levels, thereby underscoring the urgency of addressing these systemic issues [5–8].

Over the past three decades, Somalia has confronted the profound repercussions of civil war and persistent instability, severely eroding the nation's healthcare infrastructure. This prolonged conflict has propelled Somalia to some of the lowest global health indicators. Amidst these challenges, Somalia's healthcare system faces additional hindrances due to the absence of a comprehensive blood transfusion policy and the lack of accurate data on transfusion-transmissible infections (TTIs) [8–11]. To address this critical gap, this study was undertaken to identify the seroprevalence of transfusion-transmissible infections among blood donors at SOS Hospital, Heliwa-Somalia.

2. Methods

2.1. Study area

The information was retrieved from the SOS Mother and Child Hospital blood bank database. SOS is a Mother and Child Hospital with bed capacity of over 120 that serves as a cornerstone in the realm of comprehensive healthcare providing a diverse range of essential services to patients requiring assistance. Focusing on both maternal and paediatric care, the hospital delivers vital offerings, including comprehensive emergency obstetric and neonatal care (CeMONC), paediatric consultations, and admissions for communicable infections, emergency nutrition services, blood transfusion services and vaccination programs.

2.2. Study design

A retrospective descriptive analysis of blood donor data gathered at SOS Hospital from January 2016 to December 2022 was performed. The analysis focused on reviewing the monthly summaries within the blood bank for the period of the study.

2.3. Study population

The study included all individuals who volunteered to donate blood at SOS Mother and Child Hospital Blood Bank during the study period. Donors were selected based on a pre-set criterion which particularized age (16–65 years), weight (>50 kg), haemoglobin (>12 g/dl) and medical history as per the hospital protocol.

2.4. Sample size

A robust sample of over thirty-six thousand individuals who attended the SOS blood donation unit between 2012 and 2016 was meticulously gathered, ensuring comprehensive representation of the population under study. No sample size calculation was performed due to the exhaustive nature of the available dataset. The study analyzed 36,296 blood donor records, each rigorously chosen for their complete documentation of all relevant variables. Individuals with incomplete data were excluded to maintain the integrity and reliability of the analysis.

2.5. Serological investigations

Tests were conducted using serological markers of infection recommended by WHO and provided with guidelines regarding HCV and HBsAg screening employed rapid immunochromatographic kits for serum/plasma. Donors' HIV status was determined through two methods: the Determine™ HIV 1/2 and Uni-Gold™ Recombigen® HIV 1/2 test kits were previously used, while the current first-line test uses SD Bioline HIV/Syphilis Duo, capable of detecting antibodies specific to HIV-1/2 and/or *Treponema pallidum* in serum, plasma, or whole blood. The second Rapid Test for Antibody to Human Immunodeficiency Virus (HIV) utilizes a chromatographic lateral flow device with specific antigens immobilized on nitrocellulose membrane strip. Additionally, treponemal antibodies were detected using ACON rapid immunochromatographic VDRL kits, following the manufacturer's instructions.

Furthermore, SOS Hospital blood bank implemented laboratory quality assurance program where internal quality control (IQC) of serological tests done regularly, lab standard operating procedures (SOP) put in place, again SOS blood bank was part of HIV prevention, control and treatment program from WHO/MOHFGS which carry out external quality control (EQC) assessment through proficiency testing twice a year to assure competence of lab technician and institutions.

2.6. Statistical analysis

Data collected from a total of 36,296 individuals who volunteered to donate blood was meticulously organized in an Excel spreadsheet. Subsequently, the data underwent thorough cleaning and recoding processes before being subjected to detailed analysis using IBM SPSS Statistics version 25.0. The prevalence of Transfusion Transmissible Infections (TTI) was calculated by determining the number of seropositive samples per year. Meanwhile, the prevalence of anti-HIV, HBsAg, anti-HCV, and anti-TP was expressed as percentages per year. To visually represent the trends in positive cases of transfusion transmissible infections from 2016 to 2022, stacked line charts were employed. Additionally, statistical analyses involved the use of the Pearson Chi-square ($\chi 2$) test or Fisher's exact test to assess year-by-year variations in these trends. A p-value of less than 0.05 was considered statistically significant, indicating meaningful differences in the prevalence rates over the study period.

Ethics approval

Ethical clearance for this study was granted by the Institutional Review Board (IRB) of SOS College of Health Science under reference number (Ref:2023/SOS-IRB/CNS/P004). Further approval was secured from the director of SOS Mother and Child Hospital. It is important to note that this research solely utilized existing data, and participants were not directly engaged at any phase of the study. To safeguard donor confidentiality, their identities were anonymized through de-identification using specific codes, a process overseen exclusively by a proficient data specialist.

2.7. Results

According to the study, the majority of donors were male, constituting 99.8 % of the total participants, while females comprised only 0.2 %. In terms of age categories, individuals between 18 and 24 years accounted for 15.1 % of the donors, while those aged 25–44 years represented the largest group, constituting 80.4 %. Donors aged 45 and above constituted 4.5 % of the total. Regarding the type of blood donors, the majority participated as family donors, making up 80.8 % of the total, whereas voluntary donors constituted 19.2

Table 1
Demographic characteristics.

Variable	Frequency (N)	Percentage (%)
Sex		
Male	36,256	99.8
Female	40	0.2
Age categories		
18–24 years	5466	15.1
25–44 years	29182	80.4
45+ years	1648	4.5
Type of blood donors		
Family donor	29310	80.8
Voluntary donor	6986	19.2

% (Table 1).

2.8. Seroprevalence of transfusion transmissible infection

In the cohort of 36,296 donors examined, 1087 individuals (2.99 %) exhibited positive results for transfusion transmissible infections. Specifically, 725 donors (2.00 %) tested positive for HBsAg, while 10 donors (0.03 %) were identified as HCV positive. Furthermore, 335 donors (0.92 %) were syphilis positive, and 17 individuals (0.05 %) were found to be HIV positive (Table 2).

2.9. Trends of transfusion transmissible infections

The trends or pattern in the seroprevalence of HBsAg, HCV, syphilis, and HIV over the 7-year period are shown [Fig. 1]. The annual prevalence for all TTIs in 2016 was 4.276 % (190/4443), 2017 was 165 (3.59 %), 2018 was 161 (3.2 %), 2019 was 173 (3.13 %), 2020 was 140 (2.55 %), 2021 was 156 (2.56 %) and 2022 was 102 (1.98 %). The overall prevalence of TTIs seems to be declining from a high of 4.27 %–1.98 % during the seven-year period. Of the 1087 cases of TTIs detected, 725 were Hepatitis B accounting for 66.7 %, syphilis 335 (30.8 %), HIV 17 (1.6 %) and Hepatitis C 10 (0.9 %).

2.10. Seroprevalence of transfusion transmitted infections (2016–2022)

The table displays seroprevalence rates for HBsAg, HCV, syphilis, and HIV from 2016 to 2022. Hepatitis B prevalence peaked in 2016 at 2.7 % and decreased to 1.3 % in 2022. Hepatitis C rates remained consistently low, fluctuating around 0.06 %. Syphilis prevalence peaked in 2016 at 1.46 % but decreased to 0.62 % by 2022. HIV prevalence remained consistently low, with a peak of 0.09 % in 2016 and no reported cases from 2020 to 2022. Overall, there's a general trend of decline or stability in the prevalence of these infections over the observed period (Table 3).

The Chi-Square test was conducted to assess the statistical significance of the observed decline in transfusion transmitted infections (TTIs) from 2016 to 2022. The results demonstrated a highly significant association ($X2 = 57.625$, $p < 0.0001$), indicating a substantial reduction in TTIs over the specified period (Table 4).

3. Discussion

The objective of this study was to assess the seroprevalence of major transfusion-transmissible infections, including HIV, HBV, HCV, and syphilis, in a cohort of apparently healthy blood donors. The majority of the individuals who participated in this study were male, falling within the age range of 18–64 years, and were primarily donating blood for their family members.

This result aligns with other studies conducted in various countries including neighboring Ethiopia, India, and Tanzania [12–15]. The significant gender disparity among blood donors, with the majority being males, can be attributed to societal norms and cultural factors that often restrict women's participation in blood donation due to concerns related to their health and reproductive roles. The predominant type of blood donors at SOS Hospital blood banks are replacement (family) donors, which may be attributed to the obligation felt by family members to donate for their loved ones. Simultaneously, the limited number of voluntary donors may reflect a lack of community awareness regarding the importance of voluntary blood donation.

The study reveals an overall seroprevalence of transfusion-transmissible infectious agents at 2.99 % (comprising HBsAg at 2.0 %, HCV at 0.03 %, Syphilis at 0.92 %, and HIV at 0.05 %) in Mogadishu, Somalia. This study is almost comparable with the study of Chandekar et al. and Siraj et al. [16,17]. The high prevalence of hepatitis B and syphilis, compared to HIV and Hepatitis C, can be attributed to their specific modes of transmission, notably through unprotected sexual contact and mother-to-child transmission during childbirth [18,19]. This rapid spread is often exacerbated by limited awareness and healthcare accessibility, especially in regions where cultural factors influence health-seeking behaviors. People tend to have less fear of syphilis and hepatitis compared to HIV. This creates a window of opportunity for the transmission of these diseases.

Over a 7-year period, the seroprevalence of HBsAg, HCV, syphilis, and HIV decreased from 4.27 % in 2016 to 1.98 % in 2022. Among 1087 cases, Hepatitis B accounted for 66.7 %, syphilis for 30.8 %, HIV for 1.6 %, and Hepatitis C for 0.9 %. Similar decline trend was reported by several studies in the region [16,20]. The decline in the seroprevalence of transfusion-transmissible infections (TTIs) over the 7-year period could be attributed to various factors. Improved awareness and education about blood safety measures, stringent donor screening protocols, effective implementation of testing technologies, and enhanced healthcare infrastructure might

Table 2
Seroprevalence of transfusion transmitted infections.

Type of TTI	TTI status	
	Positive N (%)	Negative N (%)
Hepatitis B surface antigen (HBsAg)	725 (2.00)	35,571 (98.00)
Hepatitis B virus (HCV)	10 (0.03)	36,286(99.97)
Syphilis	335 (0.92)	35,961 (99.08)
Human immunodeficiency virus (HIV)	17 (0.05)	36,279 (99.95)
Overall TTI prevalence	**1087 (2.99)**	**35,209 (97.01)**

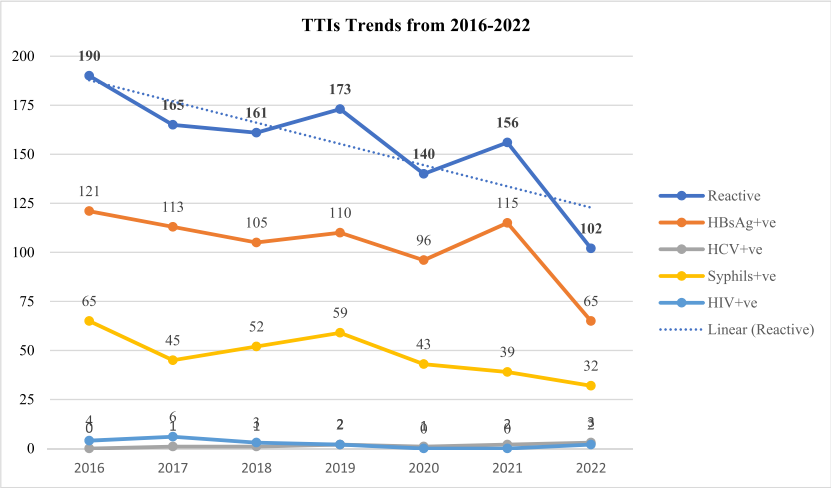


Fig. 1. Trends in the incidence of human immunodeficiency virus, Hepatitis B surface antigen, Hepatitis C virus and syphilis over the 7-year period.

Table 3
Seroprevalence of transfusion transmitted infections (2016–2022).

Year	TTI Seroprevalence			
	HBsAg + ve	HCV + ve	Syphilis + ve	HIV + ve
2016	121 (2.7 %)	0 (0.00 %)	65 (1.46 %)	4 (0.09 %)
2017	113 (2.45 %)	1 (0.02 %)	45 (0.98 %)	6 (0.13 %)
2018	105 (2.09 %)	1 (0.02 %)	52 (1.03 %)	3 (0.06 %)
2019	110 (2.0 %)	2 (0.036 %)	59 (1.07)	2 (0.04 %)
2020	96 (1.75 %)	1 (0.02 %)	43 (0.78 %)	0 (0.00 %)
2021	115 (1.9 %)	2 (0.03 %)	39 (0.64 %)	0 (0.00 %)
2022	65 (1.3 %)	3 (0.06 %)	32 (0.62 %)	2 (0.04 %)

Table 4
Seropositivity of TTIs in relation to the various years of study (2016–2022).

Years	N (%)	Transfusion Transmitted Infection Test Status		Chi square	P value
		Positive N (%)	Negative N (%)		
2016	4443	190 (17.5)	4253 (12.1)	57.625	<0.0001
2017	4595	165 (15.2)	4430 (12.6)		
2018	5029	161 (14.8)	4868 (14.8)		
2019	5511	173 (15.9)	5338 (15.2)		
2020	5479	140 (12.9)	5339 (15.2)		
2021	6095	156 (14.4)	5939 (16.8)		
2022	5144	102 (9.4)	5042 (14.3)		
Total	**36296**	**1087 (2.99)**	**35209 (97.01)**		

have contributed to this decline. The study observed a significant decline in the prevalence of TTI infections over the years. Similar trends have been observed by other studies conducted in the neighboring Ethiopia [21,22].

4. Conclusion

In conclusion, the study reveals a medium prevalence of transfusion transmitted infections with a notable decrease over time. Hepatitis B (HBsAg) stands as the most prevalent infection, succeeded by syphilis, HIV, and HCV. The gender disparity in blood donation and the dominance of replacement donors, primarily driven by societal norms restricting women's involvement and the lack of voluntary contributions, highlight the challenges in achieving a more balanced participation.

Recommendations

The study recommends implementing targeted interventions to reduce the incidence of transfusion-transmitted infections through comprehensive public health campaigns, rigorous donor screening protocols and safe blood transfusion practices. Encouraging regular,

non-remunerated volunteer blood donors would further enhance these efforts, ensuring a safer blood supply for all.

Strength and limitations

The study's strength lies in its utilization of a substantial dataset covering a seven-year period from 2016 to 2022, which is notably more extensive than that of many other studies conducted in the field in Somalia. This comprehensive dataset provides a thorough understanding of the seroprevalence and trends of transfusion-transmissible infections (TTIs) among blood donors at SOS Hospital. Despite the sizable sample size, this study had a retrospective design, restricting our ability to manage or provide information on a variety of factors that could have influenced the outcomes. This has led to some crucial variables not being included in the study like residency, culture, marital status, income, and profession. Similarly, the study's single-center approach restricts the applicability of its findings to other regions or populations within Somalia. Another limitation could be linked to the inherent drawbacks of the diagnostic tests utilized in this study, specifically the reliance on serological methods rather than nucleic acid-based techniques.

Funding

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Availability of data and materials

The data used in this study shall be available upon reasonable request from the corresponding author.

Ethics approval

Ethical clearance was obtained from the Institutional Review Board (IRB) of SOS College of Health Science with reference number (Ref: 2023/SOS-IRB/CNS/P004). Subsequent approval was also acquired from the Director of SOS Mother and Child Hospital. Notably, this research relied solely on pre-existing data, with no direct participant involvement at any point in the study. To maintain donor confidentiality, their identities were anonymized through de-identification using unique codes. The process of delinking donor identities was managed exclusively by the study's data specialist.

CRediT authorship contribution statement

Hassan Abdullahi Dahie: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Mohamed Maalim Dakane:** Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Nuradin Mohamed Mudei:** Project administration, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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